

Remote-controlled bombing

In particular, the Germans had given considerable thought to finding and hitting a target, a problem that Bomber Command had strangely neglected to take seriously. The Luftwaffe's experience in Spain had underlined the importance of night-flying and had led to the adaptation of civil radio-navigation techniques. The result was that, unlike the airships and Gothas of World War I, German night bombers did not always have to feel their way blindly through the darkness over blacked-out Britain.

All German bombers were equipped to follow a *Knickerbein* ("dog-leg") radio beam to reach their targets. The elite Kampfgruppe 100, sent in first to mark targets with incendiaries, had even more sophisticated target-finding equipment, the X-gerät and Y-gerät. The Y-gerät system, introduced in December 1940, constituted fully remote-controlled bombing. The aircraft flew along a usual beam, but reradiated the beam back to its source, so the ground station could precisely track its progress. When the bomber arrived over its target zone, it went on to automatic pilot, and as soon as it reached the map coordinates of its target, the ground station transmitted a signal that released the bomb. This technique was estimated to be able to put a bomb within a radius of 100 yards (90m), at a distance of 250 miles (400km).

Luftwaffe superiority

In practice, the Luftwaffe's radio-navigation systems did not always run smoothly. The British rapidly developed countermeasures, jamming or distorting the beams in a secret electronic war. The bombers still always preferred to attack on a clear night, when they could orient themselves by the stars and by features on the ground – the Thames Estuary proved as useful

a marker for German navigators in 1940 as it had in 1916. And, for all the technical wizardry, most bomb-aiming remained very approximate.

Whatever might be the case by day, by night the Luftwaffe had air superiority over Britain. Their fleets of 100 to 500 bombers roamed over Britain virtually unscathed. Antiaircraft batteries made a lot of noise and reassured people on the ground that something was being done, but in practice they stood little chance of hitting their targets. RAF night-interceptor squadrons operated with increasing success, especially using two-seater Bristol Beaufighters or Boulton Paul Defiants, which either flew blind, directed on to their targets by ground controllers, or used air-interception radar sets. But either way, their chances of a kill were modest. A German bomber crew, tightly packed inside the perspex nose of a Junkers, Heinkel, or Dornier over the shadowy flaming confusion of a British city, could be reasonably confident of returning home intact.



BUTTERFLY BOMB

This small antipersonnel bomb is an example of the thousands dropped by German bombers over England. It has two folding wings that revolved, slowing the rate of descent and arming the fuse.

Victory in survival

Some of the most ferocious nights of the Blitz came in April and May 1941, but this was deceptive. The Germans were by then thoroughly engaged in preparations for the invasion of the Soviet Union.

As spring turned to summer, the bombers shifted eastward and the British people, wary and watchful, slowly realized that they had come through.

London, the most frequently battered target, had been subjected to 57 nights of aerial bombardment. This had imposed a sometimes near-intolerable strain on the civilian population and on the emergency services. Yet there had been no general breakdown of society or popular pressure on the government to surrender. In the Blitz, as in the Battle of Britain, survival had amounted to a kind of victory.



SHOOTING AT V1S

British antiaircraft guns open fire at night against V1 flying bombs. The V1s were visible in the dark because of their fiery engine exhaust. Many were shot down, exploding either when hit or on impact with the ground.